

Extended hot-electron model and current bistability in insulating $a\text{-Y}_x\text{Si}_{1-x}$ filmsLe Hong Hoàng To,¹ Andrés M. Somoza,² Shamashis Sengupta,¹ Louis Dumoulin,¹ Laurent Bergé¹,¹ Stefanos Marnieros,¹ Claire Akiko Marrache-Kikuchi^{1,*}, and Miguel Ortuño^{2,†}¹Université Paris-Saclay, CNRS, IJCLab, 91405 Orsay, France²Departamento de Física—CIOyN, Universidad de Murcia, Murcia 30071, Spain

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Electron-phonon interactions play a central role in determining charge and heat transport in condensed matter systems, but in disordered and low-dimensional materials, this coupling can be strongly suppressed, leading to nontrivial transport phenomena. One striking manifestation is the appearance of bistable current-voltage (I - V) characteristics, where an imbalance between electrical heating and electron-phonon cooling yields multiple nonequilibrium steady states. In this work, we experimentally investigate this phenomenon in amorphous $a\text{-Y}_x\text{Si}_{1-x}$ thin films, where disorder can be tuned by thermal annealing. We identify the critical disorder at which bistability emerges and characterize the associated jumps in nonlinear I - V curves. To interpret these results, we develop a microscopic derivation of the phenomenological hot-electron model and extend it by incorporating electric-field-assisted hopping and phonon-mediated intergrain transitions, allowing quantitative agreement with experimental data across different disorder regimes. Complementary numerical simulations confirm the robustness of our extended framework. Our combined experimental, analytical, and numerical results provide a comprehensive description of electron-phonon decoupling and its impact on transport in disordered systems, offering a platform for studying nonequilibrium phases.

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I. INTRODUCTION

Electron-phonon interactions are fundamental to both charge and heat transport in condensed matter systems. In conventional metals, electron-phonon coupling governs resistivity and can lead to phenomena such as superconductivity. However, in disordered and low-dimensional systems, this coupling can become significantly suppressed due to weak screening and enhanced localization, which can, in turn, lead to anomalous transport behaviors, including overheating effects [1–8]. Understanding these effects is essential for interpreting transport in localized systems and developing theoretical models that capture the interplay between disorder, interactions, and phonon dynamics.

A striking example of electron-phonon decoupling in disordered systems is the appearance of bistable current-voltage (I - V) characteristics. In insulating amorphous films [1,4], InSb two-dimensional (2D) electron gases [9,10], or Josephson junction arrays [11], the imbalance between the electrical heating from the measurement and the cooling due to electron-phonon relaxation can lead to multiple steady-state solutions as the voltage bias increases. This effect is especially pronounced in materials with Arrhenius-like conductivity and is

experimentally observed through jumps of several orders of magnitude in the nonlinear I - V characteristics [12]. These nonequilibrium states give rise to hysteresis and bistability in electronic transport, highlighting a fundamental connection between electron-phonon interactions and localization.

The theoretical understanding of these phenomena was significantly advanced by Altshuler *et al.* [3], who used the so-called hot-electron model to explain the main features of the bistability observed in the experiments, particularly the jumps in the current-voltage characteristics. Further investigations were also conducted by McArdle and Lerner [12], who showed that bistability is more likely to be experimentally observable in systems exhibiting activated transport laws, and by Kasirer and Meir [13], who explained multiple jumps in the I - V curves through finite relaxation times.

In this work, we study amorphous $a\text{-Y}_x\text{Si}_{1-x}$ thin films, where disorder is controlled through thermal annealing. We were able to pinpoint the critical disorder at which the bistability emerges experimentally, thus showing that these disordered alloys provide an ideal platform to investigate the microscopic mechanisms underlying this phenomenon.

We first develop an analytical derivation that justifies the phenomenological hot-electron model and explains qualitatively the observed bistability. We then extend the model to quantitatively match our experimental results, particularly in regimes where the standard hot-electron model proves insufficient. Finally, we also perform numerical simulations that further confirm the model's validity.

The consistency between experimental results, analytical derivations, and simulations suggests that our approach forms a comprehensive picture of the role of electron-phonon interactions in disordered systems, which is applicable to a

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broad range of systems, including two-dimensional electron gases in the insulating regime. This work opens avenues for studying electronic transport in disordered materials and electron-phonon interactions therein.

II. EXPERIMENTAL SETUP

We consider an a - $\text{Y}_x\text{Si}_{1-x}$ film that has been synthesized by e-beam codeposition of yttrium and silicon under ultrahigh vacuum (the pressure during deposition was typically of a few 10^{-8} mbar) on a sapphire substrate [Fig. 1(a)]. The relative deposition rates (both on the order of $1 \text{ \AA/s}$) controlled the film composition (here $x = 14.9\%$) and were monitored *in situ* by a set of dedicated piezoelectric quartz crystals. The duration of the evaporation determined the film thickness (here $d = 350 \text{ \AA}$). Both the sample composition and thickness have been checked *ex situ* by Rutherford backscattering spectroscopy. The sample has been deposited onto a sapphire substrate and protected from oxidation by a 25-nm-thick SiO overlayer.

The samples were subsequently measured in a dilution refrigerator with a base temperature of 13 mK. Two regions of each sample (left and right) were independently probed and gave similar results. The current-voltage characteristics have been measured using a voltage bias setup, where the current was read with a Keithley 6487 picoammeter.

Heat treatments are known to increase the effective disorder level in such binary alloys [14,15]. In the present case, we measured the electronic transport properties of the as-deposited sample before annealing it under N_2 flow for 1 h at $\theta_{\text{ht}} = 80$ and 95°C , so that we were able to measure a single sample with finely evolving disorder levels. a - $\text{Y}_x\text{Si}_{1-x}$ films of similar compositions and thicknesses have been measured by transmission electron microscopy (TEM) to be amorphous and homogeneous at least up to a heat treatment temperature of 500°C [16].

III. EXPERIMENTAL RESULTS

The resistance at low-voltage bias is shown in Fig. 1(b) as a function of $1/T$ on a semilogarithmic scale. At low temperature, the resistance exhibits an activated behavior down to about 60 mK, below which it starts to saturate due to an external power input that could, for instance, be due to residual unfiltered photons heating the sample. The resistance is of the form

$$R(T) = R_0 \exp \left\{ -\frac{T_0}{T} \right\}, \quad (1)$$

where T_0 is the activation energy. The pre-exponential factor R_0 turns out to be close to the quantum of resistance $h/(2e^2)$. This kind of activated regime at low temperatures has been observed in other disordered systems [1,4–7,9,10,17–21] and in 2D systems close to a Mott's phase [22]. For systems with larger disorder, however, the activated regime has been interpreted as the manifestation of a charging energy emerging from actual or dimensionality-induced electronic granularity. Indeed, in such systems, microscopic-level disorder can lead to the formation of effective electronic grains at the mesoscopic scale. Transport then occurs through the hopping of electrons between those grains. As the heat treatment temper-

ature increases, the sample becomes more insulating, with a larger activation energy. We have moreover shown that, for films close to the superconductor-to-insulator transition, the activation energy could be enhanced due to the appearance of local superconductivity [19,20].

In Figs. 1(c)–1(f), we represent the current I as a function of the applied voltage V for two different heat treatment temperatures (80 and 95°C). Each curve corresponds to a different phonon temperature T . The voltage is scanned from negative to positive values (from left to right in the figure).

For the most insulating situation (heat treatment of 95°C), jumps of up to 4 orders of magnitude in current are observed. These jumps are all the more pronounced at low temperatures. Moreover, there is a hysteretic behavior depending on whether the applied voltage is increased or decreased. When $|V|$ decreases, the jump in current occurs at a voltage $V_<$ that is systematically lower than the voltage $V_>$ at which the jump occurs when increasing $|V|$ at the same temperature. As can be seen in Fig. 1(g), $V_>$ depends on temperature, whereas $V_<$ is almost constant for all the temperatures at which such jumps exist. This behavior has been reported in Refs. [1,3].

Figure 1 shows that smaller amplitude jumps could also be observed in some cases. These jumps are reproducible as $|V|$ is lowered, but their position could vary as $|V|$ is increased. We will comment on these later on.

In our experiment, we could additionally study the evolution of the $I(V)$ characteristics as a function of disorder. Both the ratio of the magnitude of the current jumps $I_>/I_<$ for increasing and decreasing $|V|$, respectively, and the characteristic voltages $V_<$ and $V_>$ increase with disorder. Actually, the as-deposited sample does not show any current jump at our base temperature. After a heat treatment at 80°C , small jumps are observed close to $V_> \simeq 7.5 \text{ mV}$ until $T = 40 \text{ mK}$ [Figs. 1(g) and 1(h)]. This shows how such current jumps can be induced in a single sample by progressively increasing its disorder.

IV. HOT-ELECTRON MODEL

The hot-electron model was originally developed in the study of nonlinear effects in hopping. It is a theoretical framework that emerged from a series of experimental studies on electrical and thermal properties of various materials at very low temperatures. Initially investigated by Wang *et al.* [23] in neutron-transmutation-doped Ge at 20 mK, and further developed for metals by Wellstood, Urbina, and Clarke [24], the model was subsequently extended to explain nonlinear transport in various semiconductor systems. The model's fundamental premise is that a system's resistance is determined by its electronic temperature (T_{eff}), which can significantly exceed the phonon temperature (T_{ph}) due to electron-electron interactions. When a voltage V is applied, electrons heat up and acquire an effective temperature that can be defined by comparing the corresponding resistance with the low-voltage Ohmic regime:

$$R(T_{\text{ph}}, V) = R(T_{\text{eff}}, V \rightarrow 0). \quad (2)$$

The electron-phonon coupling mechanism that relaxes the effective electron temperature back to the phonon temperature was extensively studied by Zhang *et al.* [25] in doped silicon

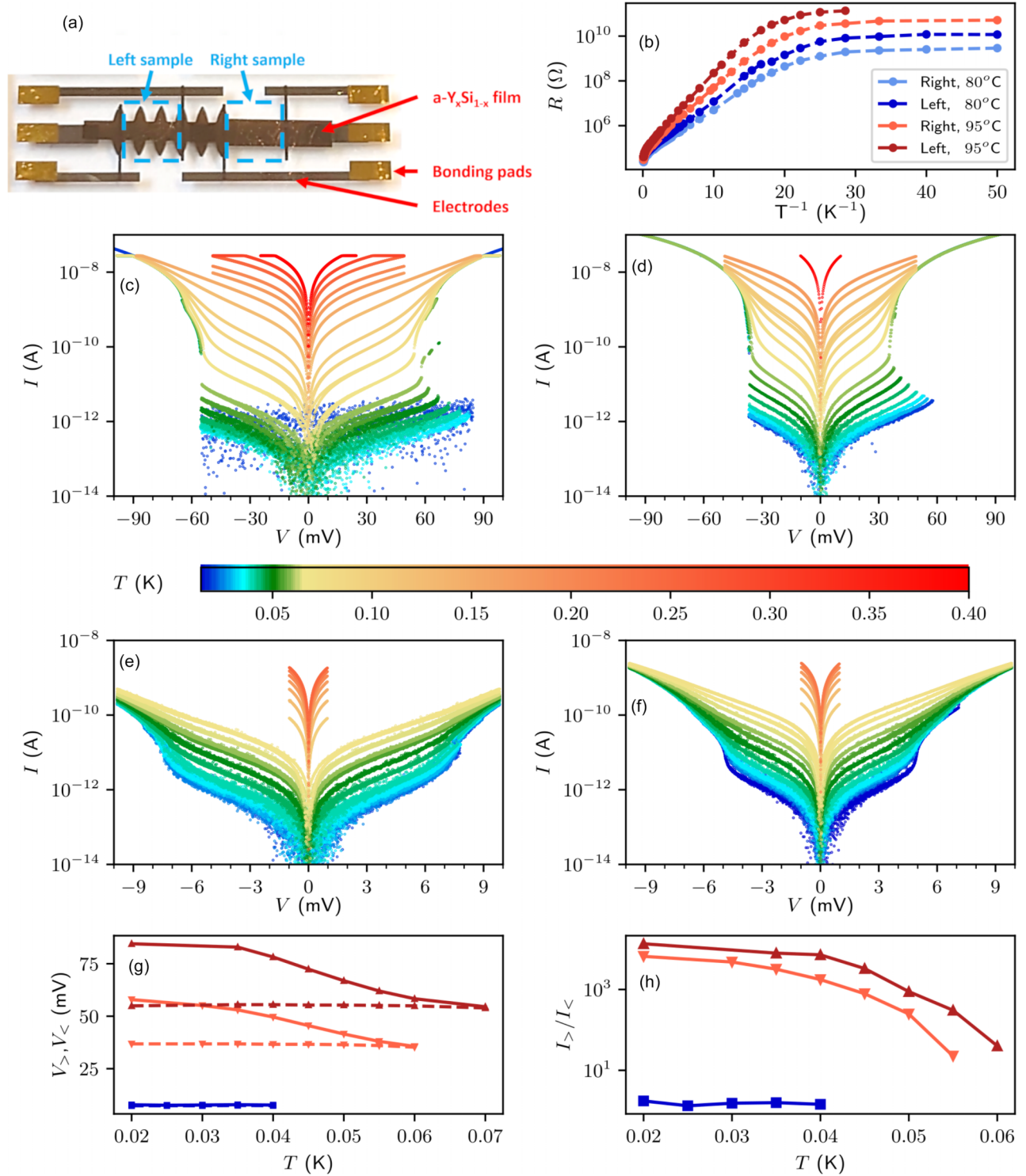


FIG. 1. (a) Picture of the sample. (b) Resistance in the linear regime (logarithmic scale) as a function of $1/T$ for all heat treatment temperatures. The points at temperatures higher than 200 mK were obtained using standard low-noise measurement techniques. At low temperature, the resistance was extracted from the $I(V)$ characteristics. Experimental I - V characteristics for the left sample at two different heat treatment temperatures: 95°C [panel (c)] and 80°C [panel (e)], and for the right sample at $\theta_{\text{ht}} = 95^\circ\text{C}$ [panel (d)] and $\theta_{\text{ht}} = 80^\circ\text{C}$ [panel (f)]. The different curves correspond to different phonon temperatures as indicated by the color scale. (g) Evolution with temperature and disorder of the critical voltages $V_>$ (full lines) and $V_<$ (dashed lines) at which the jumps occur. (h) Evolution with temperature and disorder of the magnitude of the current jumps $I_>/I_<$. The different curves in panels (g) and (h) have the same color code as in panel (b).

and germanium, and by Marnieros *et al.* [26] in the context of variable-range hopping (VRH). When $T_{\text{eff}} > T_{\text{ph}}$, electron-phonon coupling relaxes the effective electron temperature to the phonon temperature by phonon emission and absorption. The power dissipated through this process was phenomenologically found to follow the relationship

$$P = \alpha(T_{\text{eff}}^{\beta} - T_{\text{ph}}^{\beta}), \quad (3)$$

where α is a constant related to the electron-phonon coupling and, in the presence of pointlike disorder, the exponent β has been reported to be equal to 6 for systems with three-dimensional phonons and to 5 for systems with 2D phonons [27,28]. Let us note that this expression has been derived for dirty metals [23,24]. The previous expression was extended to insulators by Feigel'man and Kravtsov [28], who also derived an estimate of the coupling constant. Gershenson *et al.* [29] and Galeazzi *et al.* [30] further demonstrated the model's applicability to two-dimensional systems and strongly localized materials, showing how the hot-electron effects manifest themselves in various dimensions and disorder regimes. The dissipated power given by Eq. (3) was also observed near the superconductor-insulator transition [31–33].

For films obeying the temperature dependence of Eq. (1), equalizing the dissipated power to the electrical power supplied to the electrons, V^2/R , and taking into account Eq. (3), we infer

$$V^2 = R_0 \exp\left(\frac{T_0}{T_{\text{eff}}}\right) \alpha(T_{\text{eff}}^{\beta} - T_{\text{ph}}^{\beta}). \quad (4)$$

For phonon temperatures lower than a critical temperature given by

$$T_c = (\beta + 1)^{-\frac{\beta+1}{\beta}} T_0, \quad (5)$$

there is a voltage range for which, at a given V , Eq. (4) presents two stable solutions for T_{eff} . The lowest voltage at which such a bistability exists, which we call critical voltage, is

$$V_c = \sqrt{\frac{\alpha R_0 \beta T_0^{\beta} e^{\beta+1}}{(\beta + 1)^{\beta+1}}}. \quad (6)$$

While the critical temperature is independent of the strength of the electron-phonon coupling, the critical voltage varies as $\sqrt{\alpha}$. In the case of a nonactivated regime, i.e., when Eq. (1) is replaced by variable-range hopping, the extension of the calculation predicts a lower critical temperature T_c than in the activated case, rendering the phenomenon difficult to observe experimentally [12,28].

The phenomenological hot-electron model can explain our experimental results very well for $\theta_{\text{ht}} = 95^\circ\text{C}$, that is, for the most insulating situation. However, as we will see in the following, there is a quantitative disagreement between the model and the results for the $\theta_{\text{ht}} = 80^\circ\text{C}$ film, which calls for further theoretical developments. In Sec. V, we present a microscopic model that accounts for Eq. (3). In Sec. VI, we present an extension of the hot-electron model, which will lift the discrepancy between theory and experiment for all disorder levels.

V. MICROSCOPIC MODEL FOR THE HOT-ELECTRON MODEL

In this section, we will introduce a simple model that is capable of reproducing the hot-electron model equations introduced before [Eq. (3)].

The first ingredient we assume for this model is the existence of metallic islands within the material. This is not a strict requirement, as we will discuss in the following. Although no structural inhomogeneities were detected via TEM in our films, electronic granularity is known to play an important role in disordered systems [19,34]. This suggests the presence of regions with higher local conductivity embedded within a more insulating matrix. The observed activated dependence of the $R(T)$ reflects the contribution of these metallic regions. Indeed, disorder (impurities, defects, composition fluctuations, etc.) generates a random electrostatic potential landscape. Statistically, there will be regions where this potential is smoother than average, leading to a large localization length in that region. At the same time, a more delocalized wavefunction facilitates the screening of the electric field, thus reducing further the effective local disorder. It is therefore natural to assume that, near the metal-insulator transition, some of these regions are metallic. These regions form a percolating network at the transition, but, in the insulator, they result in metallic islands embedded in an insulating matrix. Finally, let us note that the hot-electron model has been observed to accurately describe experimental results in insulators close to the metal-insulator transition, whereas insulators deep in the insulating regime were better described by a hopping regime with electric-field effect [25]. The hot-electron model therefore seems to be valid for large localization lengths. The existence of delocalized regions has also been studied in the context of many-body localization (MBL), where they are referred to as Griffith regions. It is understood that these regions play an important role in the response of the system [35].

For simplicity, we will also assume that the metallic regions all have the same typical size and the same charging energy, due to long-range Coulomb interactions, and we will call them “grains.” As these regions are quite large, level spacing can be neglected, at least for all practical purposes.

Then, electron hops between different metallic regions can take place without the assistance of any phonon, whose transition rate can be approximated by the following:

$$\Gamma(\Delta E) \propto \int d\epsilon_1 \int d\epsilon_2 g(\epsilon_1) f(\epsilon_1) g(\epsilon_2) (1 - f(\epsilon_2)) \times \delta(\epsilon_1 - \epsilon_2 - \Delta E), \quad (7)$$

where $g(\epsilon)$ is the one-electron density of states, f is the Fermi-Dirac distribution, and ΔE is the difference in electrostatic energy between the initial and final states and/or the difference in chemical potential between the two grains. If $g(\epsilon)$ is approximately constant, the previous integral can be performed analytically [36]:

$$\Gamma(\Delta E) = \Gamma_0 \frac{\Delta E}{\exp(\Delta E/T_{\text{eff}}) - 1}. \quad (8)$$

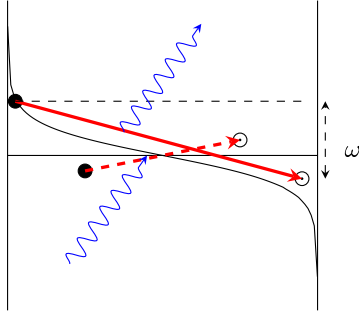


FIG. 2. Cartoon of the intragrain relaxation mechanism. An electron can jump from an occupied state to a lower-energy empty state (solid line), and the energy difference is given to an emitted phonon (blue wavy line). Alternatively, it can jump to a state with higher energy (dashed line) and, in this case, the energy gained by the electron is provided by a phonon (blue wavy line).

We assume that the system is initially neutral. Then, once a hop occurs, the two regions involved become charged at the cost of the corresponding charging energies.

Once a grain is charged, electron hops can happen easily, since the carrier does not have to pay the charging energy penalty to move to a nearby grain. As a consequence, electron hops initiated in charged grains are more probable than electron hops initiated in neutral grains. This allows us to identify the number of charged grains with the number of carriers, which is given by

$$n_c \propto \exp(-T_0/T_{\text{eff}}). \quad (9)$$

When an electric field F is applied, the electric current between two grains will be

$$I = n_c e (\Gamma(\Delta E) - \Gamma(-\Delta E)) = -\Gamma_0 \Delta E n_c e, \quad (10)$$

where $\Delta E = -e l F$ (assuming that e is the electron's charge), l being the hopping distance.

The electric power lost by this current is absorbed by the electrons, which can acquire an effective temperature higher than the phonon temperature if the electron-phonon coupling is not strong enough. We note that electrostatic interactions allow for a fast phononless thermalization mechanism between grains: One electron jumps to another state within the same grain, while another electron in a neighboring grain does the same in such a way that the total energy remains constant. As a consequence, we assume that all electrons in all grains are at the same temperature T_{eff} . This mechanism could fail if interactions are weak or if conduction is in a strong percolative regime such that most of the Joule energy is released at one point in the system. The study of Griffith regions in the context of MBL indicates that these metallic regions lead to ergodicity in $d > 1$ through the existence of avalanches [37], although the relevance of this mechanism in the dynamics of the system remains to be determined [35]. In any case, we argue that this mechanism should favor the thermalization between different grains.

The main energy relaxation mechanism responsible for the power dissipated by the electronic system then is the absorption and emission of phonons coupled to intragrain electronic transitions. A sketch of the processes is shown in Fig. 2.

The corresponding power can be obtained from the transition probability between two grain states with the emission or the absorption of a phonon, given by the integral

$$P = \Gamma_0 \int d\epsilon_1 \int d\epsilon_2 \int d\omega f(\epsilon_1)(1 - f(\epsilon_2)) \times \omega^{\beta-2} \delta(\epsilon_1 - \epsilon_2 \pm \omega) \begin{cases} 1 + N(\omega), \\ N(\omega), \end{cases} \quad (11)$$

where Γ_0 is a coupling constant, ϵ_1 and ϵ_2 are the energies of the initial and final electron states involved in the process, ω is the phonon energy, and $N(\omega)$ is the Bose-Einstein distribution. The factor $\omega^{\beta-2}$ comes from the density of phonons and from the electron-phonon coupling. The value of β will depend on the system dimensionality. The upper expression in the bracket corresponds to phonon emission, and the lower one to phonon absorption. We show in the Appendix that this integral is, to an excellent approximation, proportional to Eq. (3). Relaxation due to standard variable-range hopping produces a different expression [38].

We note that even if the low-disorder regions could not be considered truly metallic, Eqs. (7)–(10) may still remain valid. If the localization length is large, we can consider a continuous density of states. Then, within a one-electron approximation, Eq. (7) describes the transition rate of a carrier's jump. Note that the carrier cannot jump distances shorter than the localization length without the help of phonons or large many-body effects, due to orthogonality of the wavefunctions. For long hops, if an electron jumps to a neutral region, it will have to pay a charging energy proportional to the inverse of the localization length. In summary, in the absence of real metallic grains, Eqs. (7)–(11) should remain valid, and the typical hopping distance should be on the order of the localization length.

We have neglected level spacing when calculating the transition rates. However, if this were not the case, many-body processes would help reorganize the levels in such a way to reduce the effective level spacing. Furthermore, even if the dynamics require absorption or emission of a phonon, the transition rate would approximately be given by Eq. (8), as long as the level spacing is much smaller than the phonon temperature.

Our model's assumptions justify the hot-electron model, since they explain why the resistance depends only on the electron temperature. More importantly, we are able to explain in this way the phenomenological expression used in the hot-electron model [Eq. (3)]. The hypothesis that the effective grains are large also explains why this model, derived for dirty metals, is able to reproduce experimental results in disordered systems close to a conducting-to-insulating transition.

VI. EXTENDED HOT-ELECTRON MODEL

In order to fully explain the experimental results discussed in Sec. III, we have extended the hot-electron model. In Secs. VIA and VIB, we detail the two additional mechanisms we consider for electronic transport. In Sec. VIC, we develop the additional relaxation mechanisms we consider at high voltages.

A. Hopping conduction under high electric field

The first additional ingredient that we include in the model is the so-called exponential field effect in nonlinear hopping conduction [39,40]. The electric field F lowers the charging energy of the metallic regions, and the resistance should go as

$$R = R_0 \exp\left(\frac{T_0 - e l F / (2k)}{T_{\text{eff}}}\right), \quad (12)$$

where k is the Boltzmann constant. The appearance of the factor 2 will be explained in the next subsection. l is the typical hopping distance, which, for an activated behavior, should be independent of temperature and roughly equal to the grain size. For variable-range hopping, l will be T dependent (the argument of the exponential is brought to a power $n < 1$), although different models predict different temperature dependences [41,42]. The standard exponential field effect model assumes that the temperature appearing in the denominator of Eq. (12) is the phonon temperature, equal to the electron temperature. According to the hot-electron model, we assume that both temperatures do not have to be the same and, in that case, the one appearing in the denominator of Eq. (12) should be the electron temperature, i.e., the effective electronic temperature T_{eff} , in line with Eq. (2). This is the temperature that Marianer and Shklovskii consider [40], and can be significantly higher than the phonon temperature T_{ph} . The combination of the hot-electron model and Eq. (12) is able to explain most nonlinear effects we observe experimentally [25]. Let us note that it can also be used to experimentally determine the hopping distance l .

B. Tunneling contribution to conduction

The second extension we propose is to take into account conduction by tunneling. Although this mechanism is not essential to explain our experimental results, it is required to interpret our numerical simulations and may be relevant in experiments involving large values of the electric field.

The electronic conduction then is the result of a competition between the electric-field-assisted hopping mechanism described above, where an electron will hop to the conduction levels of a neighboring grain by overcoming the reduced charging energy $T_0 - e l F / (2k)$, and tunneling, where the electron can be transferred at constant energy to a grain much farther away. In the case of grains, electrons can even jump over mesoscopic distances, taking advantage of the co-tunneling mechanism [43,44]. In both cases, the transition probability decays exponentially with the distance and is proportional to

$$\exp\left(-\frac{2T_0 - e r F / k}{T_{\text{eff}}}\right) \exp\left(-\frac{2r}{\xi}\right), \quad (13)$$

where r is the hopping distance and ξ is the localization length [45]. At variance with what happens in usual variable-range hopping, the maximization of the transition probability depends on the sign of $\frac{eF}{kT_{\text{eff}}} - \frac{2}{\xi}$. For values of the electric field larger than a critical value given by

$$F_c = \frac{2kT_{\text{eff}}}{e\xi}, \quad (14)$$

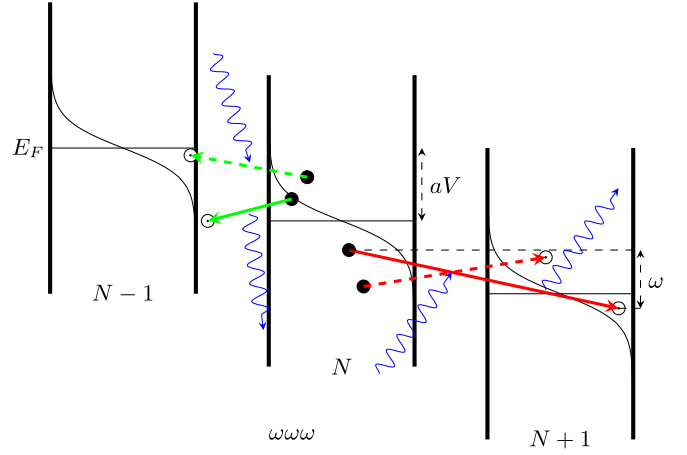


FIG. 3. Cartoon of the relaxation mechanisms between different grains. The thin black lines represent the occupation probability (Fermi-Dirac distribution) in each grain. An electron in the central grain can jump to a different neighboring grain with lower electrostatic energy (right) or to a grain with higher energy (left). In both cases, the transition can be from a high-energy occupied state to a lower-energy empty state (solid lines), and the energy difference is given to an emitted phonon (blue wavy lines) or to a state with higher energy than the initial one (dashed lines) and, in this case, the energy gained by the electron is provided by a phonon (blue wavy lines). The difference between the potential energy of two consecutive grains is aV .

the hopping distance is maximized and transport occurs by tunneling over the distance $r_F = kT_0/eF$. The resistance is then given by

$$R = R_0 \exp\left(\frac{V_0}{V}\right), \quad (15)$$

where $V = FL$ is the applied voltage and V_0 the characteristic voltage across the length L of the sample such that

$$V_0 = \frac{2kT_0L}{e\xi}. \quad (16)$$

For values of the electric field smaller than Eq. (14), activated hopping dominates and the hopping distance r is as short as possible, i.e., equal to the grain size. The resistance then reduces to Eq. (12). The behavior given by Eq. (15) was observed by Ladiou *et al.* [4].

C. Relaxation mechanisms

In addition to the intragrain relaxation (Sec. V), we have also considered a second relaxation mechanism, due to phonon-mediated charge transfers between different grains, which, as we will see, is necessary to explain quantitatively the experimental $I(V)$ curves. Some of the processes involved in this new mechanism are schematically drawn in Fig. 3. For this, we consider that electrons residing on a grain can only jump to a neighboring grain in this process.

The first possibility is that an electron from a high-energy occupied state hops to a lower empty state in a neighboring grain with lower electrical energy (red solid line). The energy difference between the states, ω , is given to an emitted phonon

(blue wavy line). The rest of the energy provided by the electric field is gained by the electrons. A second possibility is a transition to a lower-energy state in a grain with lower electrical energy (red dashed line) and the energy difference is provided by an absorbed phonon. These two processes can also take place from a grain to another neighboring grain with a higher electrical energy, represented in Fig. 3 by green lines. There are also processes to grains with similar electrical energies, not shown in the figure. In reality, the grains are not necessarily aligned along the electric field, since they are randomly distributed, but the simplified model adopted here, corresponding to a square lattice, produces good enough results.

The power exchanged with the phonons by these new processes is given by

$$P = \Gamma_0 \int d\epsilon_1 \int d\epsilon_2 \int d\omega f(\epsilon_1)(1 - f(\epsilon_2)) \times \omega^{\beta-2} \delta(\epsilon_1 - \epsilon_2 - \omega - aV) \left\{ \begin{array}{l} 1 + N(\omega), \\ N(\omega), \end{array} \right. \quad (17)$$

with the notations introduced in Sec. V, and where the energy difference between neighboring grains due to the electric field is equal to aV , where $a = le/(Lk)$, L being the size of the system in the direction of the electric field. Note that, in order not to pay an additional charging energy, the grains affected by these mechanisms are charged grains. In a manner similar to what was done in Sec. V, this integral can be approximated by (see Appendix)

$$P_2 = \alpha_2 [(T_{\text{eff}} + bV)^\beta - (T_{\text{ph}} + bV)^\beta + (T_{\text{eff}} - bV)^\beta - (T_{\text{ph}} - bV)^\beta + 2T_{\text{eff}}^\beta - 2T_{\text{ph}}^\beta] e^{-\frac{T_0 - aV}{T_{\text{eff}}}}. \quad (18)$$

The first two terms in the square brackets correspond to the processes to lower-energy grains (red lines in Fig. 3), while the third and fourth terms correspond to the processes to higher-energy grains (green lines in Fig. 3), and the last two terms to grains with the same electric energy (not shown in Fig. 3). In the parameter range of interest, the exact value of integral (17) and Eq. (18) turn out to be very similar if we choose $b = 0.32a$. Note that the energy transferred directly from the electric field to the phonons does not contribute to the electrical power given to the electrons and therefore does not heat them.

However, the new phonon-mediated charge transfer mechanisms do contribute to the current. Using the same approximation as above (see also Appendix), we obtain

$$I_2 = \alpha_2 0.42a [(T_{\text{eff}} + bV)^{\beta-1} + (T_{\text{ph}} + bV)^{\beta-1} - (T_{\text{eff}} - bV)^{\beta-1} - (T_{\text{ph}} - bV)^{\beta-1}] e^{-\frac{T_0 - aV}{T_{\text{eff}}}}. \quad (19)$$

VII. LINEAR REGIME

Let us now compare the model described in the previous sections with our experimental data, starting with the linear regime, at low V . We have already mentioned that the temperature dependence of the resistance is activated until about 60 mK, below which it starts to saturate, due to some spurious power. Within the hot-electron model, this external spurious

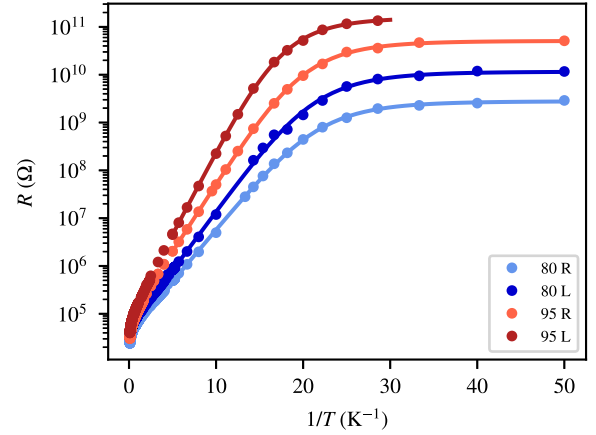


FIG. 4. Resistance in the linear regime (logarithmic scale) as a function of $1/T$ for all heat treatments. Dots correspond to the experimental results and lines to the low-temperature fitting to the hot-electron model with an extra power source (see text). The ranges of the fitting are the ranges on which the solid curves are shown and correspond to the temperatures for which I - V curves are available.

power P_{ext} , which heats the electrons, can be related to the difference between the effective electron temperature and the phonon temperature, given, in the linear regime, by

$$P_{\text{ext}} = \alpha (T_{\text{eff}}^\beta - T_{\text{ph}}^\beta). \quad (20)$$

We have extracted P_{ext} by fitting our experimental results to this model (Fig. 4), assuming that Eq. (12) is valid down to the lowest temperature for the linear regime, which is $F \rightarrow 0$. $\beta = 6$ is the value that better fits the overall results, which is consistent with what is usually found in the literature [23–26]. The agreement between experimental points and the model is quite good, indicating that the observed saturation is indeed most likely due to some parasitic power. The minimum temperature achievable then is $T_{\text{min}} = (P_{\text{ext}}/\alpha)^{1/\beta} \approx 0.047$ K. Having established the value of P_{ext} , we can now examine what our model predicts for voltages V that are beyond the linear regime.

VIII. FITTING PROCEDURE

For values of the voltage beyond the linear regime, we have fitted the experimental data of Figs. 1(c)–1(f) with the extended hot-electron model in the following way. For each experimental data point, given by V_i and I_i^{exp} , we first calculate the electrical power provided to the system through the measurement, $P = V_i I_i^{\text{exp}}$. For a given set of fitting parameters, from heat balance, we equate $P + P_{\text{ext}}$ to the power dissipated by phonons [Eqs. (3) and (18)]:

$$P + P_{\text{ext}} = \alpha (T_{\text{eff}}^\beta - T_{\text{ph}}^\beta) + P_2, \quad (21)$$

and obtain the effective temperature of the electrons. We then calculate the theoretical intensity I_i^{th} as the sum of the phononless contribution, calculated through the expression of the resistance [Eq. (12)] and of the intensity due to phonon-mediated charge transfers [Eq. (19)]:

$$I_i^{\text{th}} = \frac{V_i}{R_0} \exp\left(-\frac{T_0 - aV}{T_{\text{eff}}}\right) + I_2. \quad (22)$$

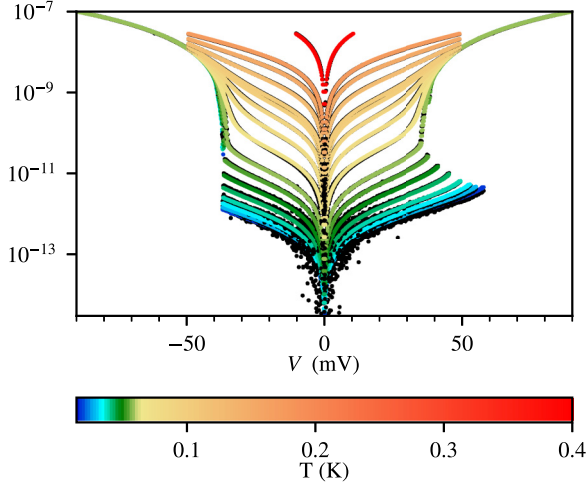


FIG. 5. Intensity (logarithmic scale) as a function of voltage for different temperatures for the right sample at a heat treatment temperature of 95 °C. Black dots correspond to experimental data and colored points to theoretical predictions.

From the difference between the theoretical and experimental intensities, we obtain the following χ^2 expression:

$$\chi^2 = \sum_i \left(\frac{I_i^{\text{exp}} - I_i^{\text{th}}}{\sigma_i^{\text{exp}}} \right)^2, \quad (23)$$

which is minimized with respect to the different parameters of the model. σ_i^{exp} is the standard deviation of the 20 experimental values around the i th one.

The free parameters for the model are the activation temperature T_0 , the resistance prefactor R_0 in Eq. (12), the strength of the nonlinear contribution a , the electron-phonon couplings α and α_2 , and the parasitic power P_{ext} . The exponent β has been taken equal to 6. Although T_0 , R_0 , and P_{ext} can be extracted independently from the linear regime, we have left them as free parameters in order to validate our fitting procedure.

IX. RESULTS

As shown in Fig. 5 for the right sample (heat treatment of 95 °C), there is an excellent agreement between the predictions of the model (colored points) and the experimental results (black dots). In this case, the second phonon-mediated relaxation mechanism is almost negligible and almost the same results would have been obtained without it. In other words, for this high level of disorder, the simple hot-electron model [Eq. (3)] is sufficient to explain the experimental data. The values found for the fitting parameters are presented in Table I. Values of T_0 , R_0 , and T_{min} obtained by this global fitting procedure are consistent with those extracted from the measurements in the linear regime (Fig. 4).

In order to assess the importance of the different relaxation mechanisms, an extremely sensitive test consists in plotting $P + P_{\text{ext}}$ as a function of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$, where T_{eff} is obtained from the resistance [Eq. (12)]. In Fig. 6, we represent this for the right sample at a heat treatment temperature of 95 °C, at different phonon temperatures. In this representation, most curves

TABLE I. Fitting parameters obtained for the right sample at the different heat treatment temperature considered. For each heat treatment temperature, the first line gives the results of the global fitting (Sec. VIII) and the second line the fit to the linear regime data (Fig. 4). Here, $T_{\text{min}} = (P_{\text{ext}}/\alpha)^{1/\beta}$ and $l = aLk/e$.

θ_{ht} (°C)	T_0 (K)	R_0 (k Ω)	T_{min} (K)	l (nm)	α (W K $^{-6}$)	α_2 (W K $^{-6}$)
80R	0.49	46.8	0.046	960	9.1×10^{-6}	4.7×10^{-4}
	0.50	39.3	0.045			
80L	0.57	42.2	0.047		2.8×10^{-6}	1.2×10^{-3}
	0.57	47.8	0.046			
95R	0.65	79.7	0.048	160	4.0×10^{-5}	1.5×10^{-5}
	0.65	77.3	0.049			
95L	0.77	100.6	0.053	130	2.5×10^{-5}	6.9×10^{-5}
	0.79	85.7	0.055			

are almost perfectly linear, with a slope of 1 in double logarithmic scale, indicating that the standard intragrain relaxation mechanism, as described by the usual hot-electron model and given by Eq. (3), is by far the dominant one. Actually, including the effect of the electric field on the hopping conduction [Eq. (12)] yields similar results, since the contribution of the field is negligible.

The sample with a heat treatment at 80 °C is less insulating than the one at 95 °C and presents a much less pronounced bistability. In Fig. 7, we show the I - V curves for the right sample (heat treatment of 80 °C). Colored points correspond to the predictions of the hot-electron model, while black dots to the experimental results. We have plotted a smaller region than in Fig. 1 in order to appreciate better the quality of the fit. Although the main trends of the experimental results are reproduced, there are clear discrepancies between experimental points and these theoretical predictions. The predictions of the hot-electron model, even when including the electric-field effect on hopping conductivity [Eq. (12)], cannot reproduce

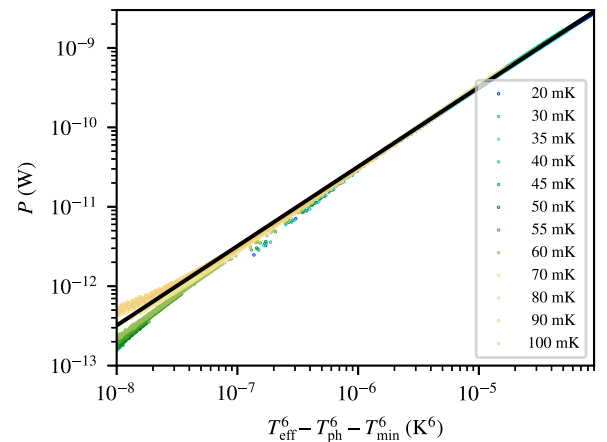


FIG. 6. $P + P_{\text{ext}}$ as a function of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$ on a double logarithmic scale for the right sample annealed at a heat treatment temperature of 95 °C and for $\beta = 6$. Here, P is the experimental electrical power, and T_{eff} is extracted from the value of the resistance [Eq. (12)]. The full black line has a slope of 1 and corresponds to the standard intragrain relaxation mechanism [hot-electron model; Eq. (3)].

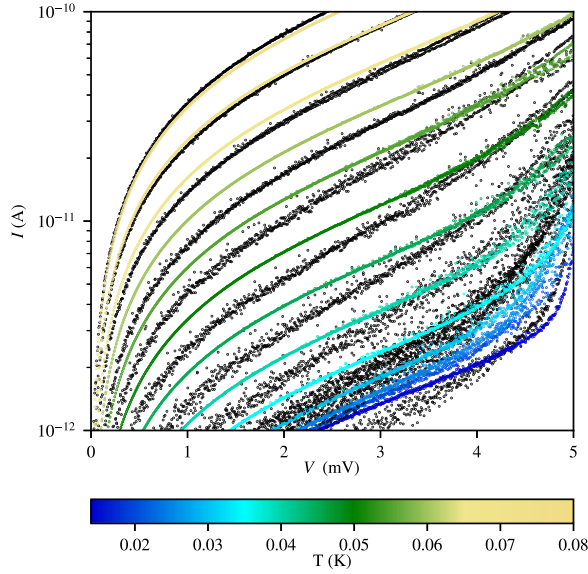


FIG. 7. Intensity (logarithmic scale) as a function of voltage for different temperatures for the right sample at a heat treatment temperature of 80 °C. Black dots correspond to experimental data and colored points to theoretical predictions using the hot-electron model.

the experimental I - V curves for this situation, no matter what parameters of the model we use, unless high values of β are employed (typically, having β as a free fitting parameter would give $\beta \simeq 8$). Most importantly, we also note that the parameters obtained with the previous fit to the hot-electron model ($T_0 = 0.40$ K and $R_0 = 150.0$ k Ω) drastically differ from their values obtained from the linear regime.

Another way of assessing that the hot-electron model does not adequately describe the experimental data for $\theta_{\text{ht}} = 80$ °C is to examine the $P + P_{\text{ext}}$ plot as a function of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$. As seen in Fig. 8, the curves are not linear. More importantly,

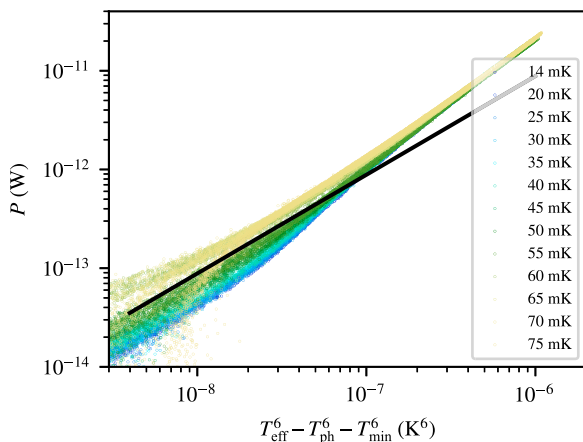


FIG. 8. $P + P_{\text{ext}}$ as a function of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$ on a double logarithmic scale for the right sample at a heat treatment temperature of 80 °C and for $\beta = 6$. Here, T_{eff} is obtained from the resistance [Eq. (12)]. The full black line has a slope of 1 and corresponds to the standard intragrain relaxation mechanism [hot-electron model; Eq. (3)].

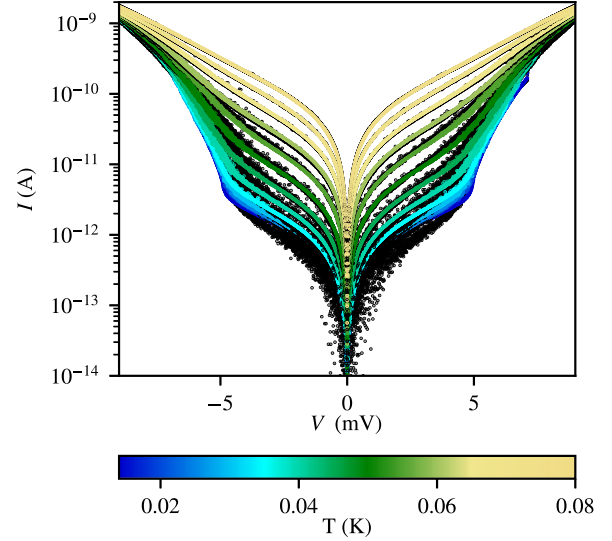


FIG. 9. Intensity (logarithmic scale) as a function of voltage for different temperatures for the right sample at $\theta_{\text{ht}} = 80$ °C. Black dots correspond to experimental data, and colored points to theoretical predictions.

at large voltages (large values of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$), the curves tend to collapse onto a line of slope larger than 1, corresponding to a value of $\beta \simeq 8$. There is therefore a strong discrepancy between the experimental data and the prediction of the phononless hot-electron model. To lift this discrepancy, one can either employ the hot-electron model with a temperature-dependent β , on the order of 8 at temperatures of the order of 200 mK, or consider the additional phonon-mediated relaxation mechanism described in Sec. VIC. We considered the first option to be less justified (we comment on this later) and will focus on the second in what follows.

As can be seen from Fig. 9, the extended hot-electron model, which considers the additional electric-field effect on the hopping conduction and the phonon-mediated charge transfer and relaxation mechanisms, reproduces extremely well (colored dots) the experimental data (black dots) for $\theta_{\text{ht}} = 80$ °C, especially in the intermediate voltage region (Fig. 10), which was not well described by the hot-electron model (Fig. 7). The values of the fitting parameters are given in Table I, and one can appreciate the good agreement of these values with those obtained from the linear regime.

It is interesting to compare the phononless and the phonon-mediated relaxation mechanisms. This is what we do in Fig. 11, where we plot the total power provided to the electron system as a function of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$ on a double logarithmic scale. In this case, T_{eff} is obtained from equating the power provided and the power dissipated by the electron system. At low voltages, the standard dissipation mechanism dominates, while at high voltages the new mechanism is up to five times larger than the standard one.

The first mechanism is valid for all grains, for which the main contribution comes from neutral grains, just because there are more of those. The second mechanism is dominated by the contribution by charged grains.

A surprising result is that the value of the electron-phonon coupling of the second (intergrain) mechanism α_2 is much

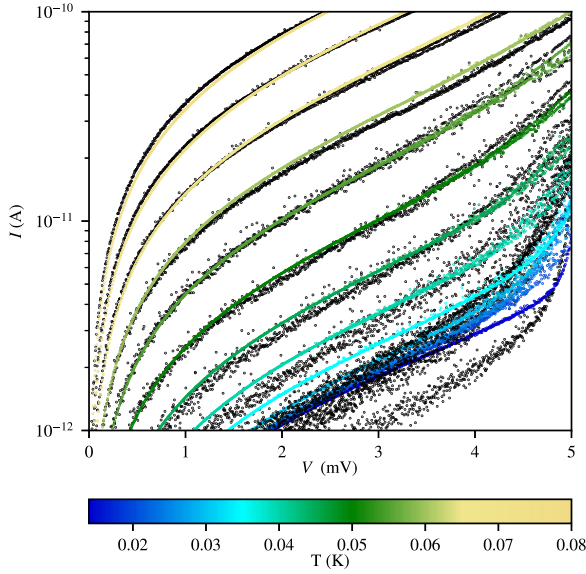


FIG. 10. Intensity (logarithmic scale) as a function of voltage for different temperatures for the right sample at $\theta_{\text{ht}} = 80^\circ\text{C}$. Black dots correspond to experimental data, and colored points to theoretical predictions using the extended hot-electron model. The scale is the same as in Fig. 7.

higher (about a factor of 50) than the original coupling α , whereas one would expect $\alpha > \alpha_2$, since α depends on the coupling of wavefunctions in the same grain, while α_2 depends on the coupling of wavefunctions in different grains. This discrepancy could be due to a larger extension of the wavefunctions of the charged grains, as compared to the wavefunctions of the neutral grains.

X. NUMERICAL SIMULATIONS

A. Numerical procedure

In order to validate our theoretical derivation and in addition to the experiments described above, we conducted

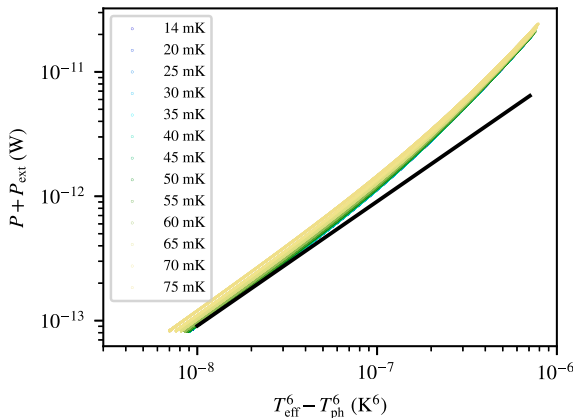


FIG. 11. $P + P_{\text{ext}}$ as a function of $T_{\text{eff}}^\beta - T_{\text{ph}}^\beta$ on a double logarithmic scale for the right sample at $\theta_{\text{ht}} = 80^\circ\text{C}$ and for $\beta = 6$. Here, P is the experimental electrical power, and T_{eff} is extracted from Eq. (21). The black line has a slope equal to 1 and corresponds to the hot-electron model, including Eq. (12).

numerical simulations. We consider the simplest model that produces an activated behavior and incorporates the ingredients of the hot-electron model. We assume sites arranged on a square lattice of size $L \times L$ characterized by an integer charge n_i at each site i , a charging energy E_c , assumed to be the same for all sites, and a given electron temperature T_{eff} , also assumed to be uniform across the sample. Each site represents a metallic region.

Carriers hop from site to site, transferring a quantized unit charge. The energy required to hop from a site i to another site j is

$$\Delta_{i,j} = E_c((n_j + 1)^2 - n_j^2 + (n_i - 1)^2 - n_i^2) - (x_j - x_i)F, \quad (24)$$

where x_i is the coordinate of site i along the direction of the electric field F .

If the level spacing within a site is much smaller than the other relevant energies of the system, which will be true if the metallic regions represented by the sites are large enough, as discussed above, the phononless transition rate between grains, assumed to be at the same effective temperature T_{eff} , can be approximated by

$$\Gamma_{i,j} = \Gamma_0 \frac{\Delta_{i,j}}{e^{\Delta_{i,j}/T_{\text{eff}}} - 1} \exp\{-2r/\xi\}, \quad (25)$$

where we have used Eq. (8) and the exponential dependence on distance comes from the matrix elements of the wavefunctions on different sites.

The electrostatic energy involved in such a hop is gained by the electron, which ends up in a high-energy state. Electron-electron interactions distribute this extra energy between all the electrons, establishing a well-defined effective temperature that depends on the specific heat of the grains. Electrons then dissipate this energy to phonons according to Eq. (3).

The model is easily generalizable to the case where the effective temperature is not the same in all grains. The energy gained could be deposited in the grains involved in a jump and distributed slowly to the rest of the grains. This slow thermalization may be related to the existence of secondary jumps, mentioned in Sec. III.

B. Numerical results

We have considered square samples with dimensions $L = 40$ and a localization length $\xi = 1$, in units of the lattice spacing. The charging energy is taken equal to unity, establishing our unit of energy, and the transition rate, Γ_0 , is also taken equal to 1, establishing our unit of time. Our simulations in the linear ($V \rightarrow 0$) regime produce an activated behavior with an activation energy $T_0 = E_c = 1$ (in units of $k_B = 1$), as expected, and a prefactor $R_0 = 3.13$ (in units of $h/e^2 = 1$). This is consistent with our previous results [19].

For our simulations, we have chosen two values for the electron-phonon coupling, $\alpha = L^2$ and $5L^2$. The value of α controls by how much the effective temperature differs from the phonon temperature for a given supplied electrical power. The I - V curves for these two values of the electron-phonon coupling are shown in the top panels of Fig. 12. The left panel corresponds to $\alpha = L^2$ and the right panel to $5L^2$. The voltage is scanned from large negative values to positive values, so

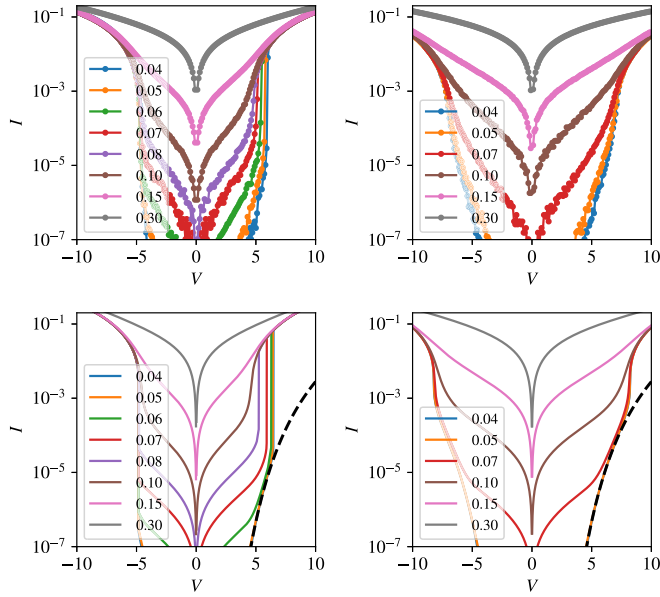


FIG. 12. I - V curves as given by our numerical simulations (top panels) for two values of the electron-phonon coupling $\alpha = L^2$ (left) and $5L^2$ (right) for the temperatures (in units of T_0) shown in the figure. The bottom panels correspond to numerical results of the extended hot-electron model for the same parameters as in the simulations. The dashed black line is the tunneling contribution.

that the left part of each panel corresponds to decreasing values of $|V|$ and the right part to increasing values. For the weakest electron-phonon coupling $\alpha = L^2$, there is a drastic bistability and hysteretic effects, while both effects disappear in the second case at stronger α .

In the bottom panels of Fig. 12, we present for comparison the results of the extended hot-electron model for the same parameters as used in the simulation. Let us emphasize that, unlike what we showed in the previous section, Fig. 12 is not a fit. T_0 and α are known from the outset, and R_0 is fitted from the simulations in the linear regime. The only unknown parameter is the strength $el/(2k)$ of the nonlinear term in Eq. (12), which corresponds to the length of the typical jump. We have chosen $el/(2k) = 1.7$, which is the value that best describes our simulation results. The vertical lines in Fig. 12 correspond to the points where the I - V curves present an infinite slope, and experimentally, the jumps between the two branches of the curves are expected to occur before these points are reached due to inevitable small fluctuations in the experimental conditions. Taking all this into account, we note that the agreement between the simulations and the model is quite good, signifying that the extension of the hot-electron model we propose very likely contains all the relevant ingredients.

The dashed lines in the bottom panels of Fig. 12 correspond to the contribution of tunneling, as described by Eq. (15). We see that this effect contributes to the disappearance of the bistability, since the I - V curves cannot go below the tunneling curve. An increase in the electron-phonon coupling produces an increase in the critical voltage for the jumps, which results in a greater importance of the nonlinear electric-field term and of the tunneling contribution.

XI. DISCUSSION

Before concluding, let us finish by a few comments. First, let us note that nonlinear effects in hopping systems are often described as due to two mutually exclusive explanations, the hot-electron model and the so-called exponential field-effect model, which considers that the resistance exponentially depends on the electric field [Eq. (12)]. Incorporating this field-effect dependence into the hot-electron model, as done in this work, can explain most experimental results [25] within a single consistent model.

The applied electric field then lowers the activation energy required for electron hopping. In practical terms, this leads to a field-dependent resistance given by an exponential term involving the field strength and the hopping distance, which provides a possible tool for estimating the localization length, which should be on the order of the hopping distance, without involving the dielectric response, which is usually difficult to calculate. We found the localization length, $\xi \simeq l$, to be relatively large, on the order of a few hundreds of nanometers (see Table I), indicating that electronic wavefunctions extend over many sites with significant overlap, thereby making standard variable-range hopping models—which imply impurity states—insufficient to accurately describe conduction in this regime. The values of ξ obtained with this approach are consistent with those obtained from equating the activation energy to a charging energy:

$$kT_0 = \frac{e^2}{4\pi\epsilon_0\epsilon_r\xi}. \quad (26)$$

For example, inserting in this equation the values of T_0 and $\xi \approx l$ for the heat treatment at 80 °C, we obtain a relative dielectric constant of $\epsilon_r = 35$, which is of the expected order of magnitude.

Second, we note that the metallic grains that we have considered in the model developed here have been predicted in other contexts. For example, phase coexistence is expected to result from the first-order Mott metal-insulator transition in dilute 2D electron gases at semiconductor interfaces, in quasi-2D Mott organics, or in transition-metal dichalcogenide moiré bilayers [22,46,47].

It may seem surprising that, despite the system's high inhomogeneity, electrons from different metallic islands thermalize to a single, well-defined temperature. We hypothesize that this uniform thermalization is the result of many-body quantum effects and interactions, drawing a parallel to the phenomenon of MBL. In the context of MBL, a similar scenario is predicted near the phase transition, where ergodic regions (thermally coupled) coexist with localized regions (thermally isolated). We believe, although this is speculative, that an “avalanche instability” may facilitate the system's thermalization, as localized thermal “bubbles” act as thermal baths, enabling heat transport throughout the system.

Third, we note that the drastic increase in the intensity observed beyond a critical voltage can be triggered independently by either an increase in effective temperature or high voltages [9,10,48,49].

Fourth, in the framework of the present model, the so-called activationless VRH (i.e., temperature-independent conductivity) observed in some systems under large electric field [50] can be understood as resulting from the effective

electronic temperature T_{eff} being independent of the phononic temperature at high voltages.

Fifth, phononless hopping has been invoked to explain the existence of a universal conductivity prefactor in the resistance of 2D systems [51]. Our experimental results also show prefactors close to the quantum of resistance, which is yet another indication that phononless hopping is the main mechanism at play here.

Finally, we have also tried to fit the results for the samples annealed at 80 °C by allowing the exponent β to change. Our fitting procedure considering β as a free parameter gives results of similar quality to those presented here, although some of the parameters seem to be less reasonable than the one found with the extra phonon relaxation mechanism. In this analysis, the best fitted value of β is very close to 8, which could be due to relaxation by phonons involving two electron jumps, due to interactions, thus contributing to two more integrals in the calculation of the dissipated power, and therefore to an extra factor ω^2 (see Appendix).

XII. CONCLUSION

In this paper, our aim was to provide both a microscopic foundation and necessary extensions to the hot-electron model to fully account for the experimental observations of bistable current-voltage (I - V) characteristics across different disorder regimes in insulators.

In activated insulators, electronic transport occurs through hops between metalliclike regions, or effective grains. Coulomb interactions ensure phononless thermalization throughout the system and justify the existence of a well-defined electron temperature T_{eff} . The dominant energy relaxation mechanism is then related to intragrain electronic transitions and results in the hot-electron model. Our results show that this is the main process at play for our more insulating samples—though still relatively close to the metal-to-insulator (or superconductor-to-insulator) transition.

Closer to the transition, where the conductivity is larger, we have extended the hot-electron model to include electric-field-assisted hopping and phonon-mediated charge transfers between grains, which introduce additional relaxation pathways. These mechanisms allow the model to accurately capture the full range of nonlinear I - V characteristics, particularly in the intermediate disorder regime where the standard hot-electron model alone fails.

Beyond providing a quantitative description of nonlinear transport, these studies open a window onto the fundamental physics of disordered electronic systems. In particular, they touch onto the topic of nonequilibrium phases, such as many-body localized states, and are important to deepen our understanding of thermalization and ergodicity breaking in interacting systems. On a practical level, such insights may be applied to design novel electronic devices exploiting bistability, weak electron-phonon coupling, or controlled switching between conduction states, with potential applications in, for instance, hot-electron bolometers and quantum sensing [52].

ACKNOWLEDGMENTS

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DATA AVAILABILITY

There are no publicly available research data or software supporting this manuscript. Requests for further information or data should be sent to the authors.

APPENDIX: TRANSITION RATES

First, let us solve the equation for the dissipated power in the case of the standard intragrain relaxation process [Eq. (11)]:

$$P = \Gamma_0 \int d\epsilon_1 \int d\epsilon_2 \int d\omega f(\epsilon_1)(1 - f(\epsilon_2)) \times \omega^{\beta-2} \delta(\epsilon_1 - \epsilon_2 \mp \omega) \left\{ \frac{1 + N(\omega)}{N(\omega)}, \right. \quad (\text{A1})$$

where the upper term in the bracket and the $-\omega$ term correspond to the emission of a phonon, whereas the lower term in the bracket and the $+\omega$ term correspond to the absorption of a phonon. If the electrons are at the same temperature T_{eff} , the integrals over ϵ_1 and ϵ_2 can be directly calculated:

$$F(\pm\omega) = \int d\epsilon_1 \int d\epsilon_2 f(\epsilon_1)(1 - f(\epsilon_2)) \delta(\epsilon_1 - \epsilon_2 \mp \omega) = \pm \frac{\omega}{e^{\pm\omega/kT_{\text{eff}}} - 1}. \quad (\text{A2})$$

Then, the emitted power is given by

$$P_e = \Gamma_0 \int d\omega F(\omega) \omega^{\beta-2}. \quad (\text{A3})$$

As we will see, the result of this integral for the relevant range of parameters is, to an excellent approximation, proportional to

$$P_e \propto T_{\text{eff}}^\beta. \quad (\text{A4})$$

In the most general case, where there is a possible difference in electrostatic energy between the two states involved in the intergrain process, the power emitted to the phonons is given by

$$P_e = \Gamma_0 \int d\epsilon_1 \int d\epsilon_2 \int d\omega f(\epsilon_1)(1 - f(\epsilon_2)) \times \omega^{\beta-2} \delta(\epsilon_1 - \epsilon_2 - \omega - aV)(1 + N(\omega)), \quad (\text{A5})$$

where aV is the electrostatic energy difference. Then, the emitted power is given by

$$P_e = \Gamma_0 \int d\omega F(\omega + bV) \omega^{\beta-2}(1 + N(\omega)). \quad (\text{A6})$$

The result of this integral for the relevant range of parameters is, to an excellent approximation, proportional to

$$P_e \propto (T_{\text{eff}} - caV)^\beta, \quad (\text{A7})$$

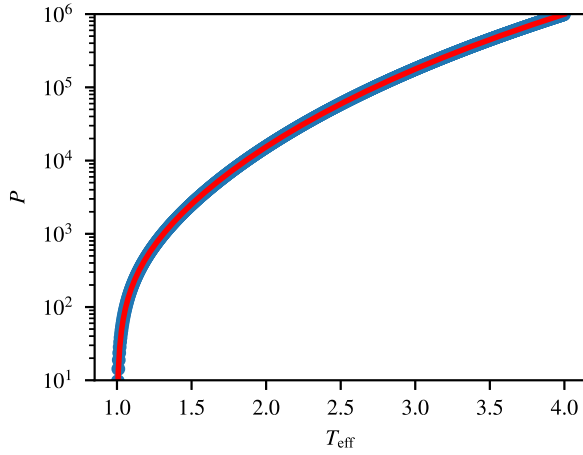


FIG. 13. Power as a function of effective temperature in units of the phonon temperature for an energy difference of $0.5kT_{ph}/e$. The continuous curve corresponds to the exact result and the dots to the approximation given by Eq. (A8).

where c is a constant that we will determine by fitting the exact result to this expression.

Assuming that the grains are arranged in a square lattice and that the applied electric field is along one of the direction between adjacent grains, the total emitted power is the sum of four terms of the form (A5), with electrostatic energy differences: aV , $-aV$, and 0 (twice). The absorbed power has similar contributions as the emitted one, with the phonon temperature playing the role of the effective temperature. The emitted power mainly depends on the effective electron temperature T_{eff} through the dependence of $F(\omega)$, with $(1 + N(\omega)) \simeq 1$ in Eq. (A1). On the contrary, the absorbed power depends both on the phonon temperature T_{ph} and on T_{eff} , via the dependence in $N(\omega)$ in Eq. (A1) and in $F(\omega)$. However, to an excellent approximation, the difference of the emitted and absorbed powers is given by

$$P = c[(T_{eff} - bV)^\beta + (T_{eff} + bV)^\beta + 2T_{eff}^\beta - (T_{ph} - bV)^\beta - (T_{ph} + bV)^\beta - 2T_{ph}^\beta]. \quad (A8)$$

In Fig. 13, we represent the exact result of the dissipated power (dotted curve) and the approximation (continuous curve) [Eq. (A8)] for an energy difference of $0.5kT_{ph}/e$, which is a typical large value for the voltages considered. The best

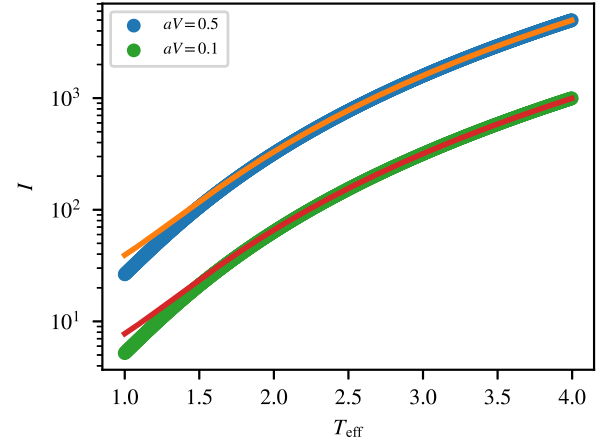


FIG. 14. Intensity as a function of effective temperature in units of the phonon temperature for two values of the electric potential, $0.5kT_{ph}/e$ and $0.1kT_{ph}/e$. The symbols correspond to the exact result and the continuous curves to the approximation given by Eq. (A9).

fit is for $c = 122$ and $b = 0.16a$. We note the excellent agreement between both curves.

The intensity is given by equations similar to that for the power, differing in that there is a power of ω less in the integrand and that the signs of the different contributions depend on whether the transition is in the direction of the field or opposite to it, instead of whether the phonon is absorbed or emitted. To a relatively good approximation, the intensity is given by

$$I = d[(T_{eff} - bV)^{\beta-1} - (T_{eff} + bV)^{\beta-1} + (T_{ph} - bV)^{\beta-1} - (T_{ph} + bV)^{\beta-1}]. \quad (A9)$$

In Fig. 14, we represent the exact expression (symbols) and the approximated one (continuous curves) [Eq. (A9)] as a function of the effective temperature for a large voltage, $0.5kT_{ph}/e$, and a small one, $0.1kT_{ph}/e$. The fitted value of the parameter is $d = 24.4$ and b is the same as for the power. At low values of T_{eff} , the agreement between both curves is not so good, but the corresponding contribution is so small that this disagreement has no practical effect. The total contribution to the intensity of this phonon-assisted hopping is small, since the two main contributions [the first two terms in Eq. (A9) involving T_{eff}] almost cancel each other, unlike for the power, where these two contributions add up.

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